b) $n < x$ if and only if $n < \lceil x \rceil$.

Soln: Let, $n < x$. By Property 2 from Table 1, $x \leq \lceil x \rceil$.
$\therefore n < \lceil x \rceil$.

Let, $n < \lceil x \rceil$. Let, $\lceil x \rceil = m$. By Defn. 12, m is an integer
By Property (1b) from Table 1, $m-1 < x \leq m$.
$\because x$ is a real no. $\therefore x < m$. $\because n < m$ $\therefore n \leq m-1$
$\therefore n < x$ □

54) Show that if x is a real number and n is an integer, then
a) $x \leq n$ if and only if $\lceil x \rceil \leq n$.

Soln: Let, $x \leq n$. Let, $\lceil x \rceil = m$. By Defn. 12, m is an integer.
$\therefore m-1 < x \leq m$, by Property 1(b).
Given, x is a real no. $\therefore x < m$. By Defn. 12, m is the smallest integer larger than or equal to x. Since, n is arbitrary, $m \leq n \Rightarrow \lceil x \rceil \leq n$.
Let, $\lceil x \rceil \leq n$. By Property 2 from Table 1, $x \leq \lceil x \rceil$. $\therefore x \leq n$. □

b) $n \leq x$ if and only if $n \leq \lfloor x \rfloor$.

Soln: Let $n \leq x$. By Defn. 12, $\lfloor x \rfloor$ is the largest integer less than or equal to x. Since n is arbitrary, so $n < \lfloor x \rfloor$.
Let, $n < \lfloor x \rfloor$. By Property 2 from Table 1, $\lfloor x \rfloor \leq x$. $\therefore n \leq x$ □

55) Prove that if n is an integer, then $\lfloor n/2 \rfloor = n/2$ if n is even and $(n-1)/2$ if n is odd.

Soln: Suppose n is an even integer. By Defn. 1 from Sec-1.7, $\exists K \in \mathbb{Z}$ s.t. $n = 2K$. $\lfloor n/2 \rfloor = \lfloor 2K/2 \rfloor = \lfloor K \rfloor = K$ ($\because K$ is an integer)
$= n/2$

Suppose n is an odd integer. By Defn. 1 from Sec-1.7, $\exists K \in \mathbb{Z}$ s.t. $n = 2K+1$
$\lfloor n/2 \rfloor = \lfloor (2K+1)/2 \rfloor = \lfloor K + 1/2 \rfloor = \lfloor K + 0.5 \rfloor = K = (n-1)/2$ □